SWID Data Glossary

Key Terms and Definitions for the Solavise Women in Data Programme

This glossary covers the most important words and ideas you will come across in your data journey. Refer to it any time you meet a term you are not sure about. Terms are grouped by topic so it is easy to find what you need.

General Data Concepts

Dataset	A collection of related information organised in rows and columns. Think of it like a spreadsheet full of records.
Data	Raw facts and numbers that have not yet been interpreted or analysed.
Database	A structured system for storing and managing large amounts of data, usually accessed using special software.
Row	A single record in a dataset. Each row represents one item, person, or event.
Column	A single field or attribute in a dataset. Each column holds one type of information, such as names or ages.
Variable	Another word for a column. A variable holds one specific type of information about each record.
Observation	One row of data. It represents a single instance or measurement in your dataset.
Feature	A measurable property used to describe each record. In a student dataset, age and score are both features.
Structured Data	Data that is organised in a fixed format with rows and columns, like a table or spreadsheet.
Unstructured Data	Data that does not have a fixed format, such as emails, images, or social media posts.
Categorical Data	Data that represents groups or labels rather than numbers. Examples include gender, country, or product type.
Numerical Data	Data that is expressed as numbers and can be measured or counted. Examples include age, price, and temperature.
Discrete Data	Numerical data that can only take specific separate values, such as the number of students in a class.
Continuous Data	Numerical data that can take any value within a range, such as height or weight.
Primary Key	A unique identifier for each row in a dataset. It makes sure no two rows are identical.
Sample	A smaller group of data taken from a larger dataset to represent the whole.
Population	The full group that data is being collected about. A sample is taken from the population.
Data Type	The kind of value stored in a column, such as text (string), whole number (integer), decimal (float), or date.
CSV File	A common file format where data is saved as plain text and values are separated by commas. Short for Comma Separated Values.
DataFrame	A table of data in Python with rows and columns, created using the Pandas library. It is the main structure you will work with.

Statistics and Math

Mean	The average of a set of numbers. You add all the values together and divide by how many there are.
Median	The middle value when all numbers are arranged in order. It is less affected by very high or very low values than the mean.
Mode	The value that appears most often in a dataset. A dataset can have more than one mode.
Range	The difference between the highest and lowest values in a dataset.
Standard Deviation	A measure of how spread out the values in a dataset are from the mean. A small value means most values are close to the mean.
Variance	The average of the squared differences from the mean. Standard deviation is the square root of variance.
Distribution	The way values are spread across a dataset. It shows how often each value appears.
Normal Distribution	A bell shaped pattern where most values cluster around the mean and fewer values appear at the extremes. Very common in real world data.
Skewness	A measure of how lopsided a distribution is. Positive skew means the tail is on the right. Negative skew means the tail is on the left.
Outlier	A value that is very different from the rest of the data. It can be unusually high or unusually low.
Percentile	A value below which a certain percentage of data falls. The 50th percentile is the same as the median.
Quartile	One of three points that divides data into four equal parts. Q1 is the 25th percentile, Q2 is the median, and Q3 is the 75th percentile.
Interquartile Range	The range between Q1 and Q3. It covers the middle 50 percent of the data and is used to spot outliers.
Correlation	A measure of how strongly two variables move together. A value close to 1 means a strong positive link. Close to 0 means little or no link.
Frequency	How many times a particular value appears in a dataset.
Probability	A number between 0 and 1 that shows how likely something is to happen. 0 means impossible and 1 means certain.
Statistical Bias	A consistent error in data collection or analysis that causes results to be systematically wrong in one direction.
Hypothesis	A testable statement or prediction about your data. Data analysis is often used to check whether a hypothesis is true.
Null Hypothesis	The default assumption that there is no effect or no relationship between two things in your data.
p value	A number used in statistics to decide whether a result is significant. A p value below 0.05 usually means the result is not due to chance.

Python and Pandas

Python	A popular programming language used widely in data analysis, machine learning, and automation because it is easy to read and write.
Library	A collection of pre written code that you can import into your Python programme to save time. Pandas and NumPy are examples of libraries.
Pandas	A Python library used for working with data in table form. It lets you load, clean, filter, and analyse datasets easily.
NumPy	A Python library for working with numbers and arrays. It makes mathematical operations on large sets of numbers very fast.
Series	A single column of data in Pandas. It is like a list of values with a label for each one.
DataFrame	A two dimensional table in Pandas with rows and columns, similar to a spreadsheet.
Index	The row labels of a DataFrame. By default, rows are numbered starting from 0.
Method	A built in action you can perform on a DataFrame or Series. For example, .head() and .describe() are methods.
Function	A block of reusable code that performs a specific task. You can use built in functions or write your own.
Import	The command used to load a library into your programme. For example: import pandas as pd.
Variable	A name you give to a piece of data in Python so you can use it later in your code.
Loop	A way to repeat a block of code multiple times, usually going through each item in a list or column.
Conditional	A piece of code that runs only when a certain condition is true. It uses if, elif, and else statements.
Boolean	A value that is either True or False. Used for filtering and conditions in Python.
Filter	A way to select only the rows in a DataFrame that meet a certain condition.
GroupBy	A Pandas method that groups rows together based on the values in one or more columns so you can calculate summaries for each group.
Merge	A way to combine two DataFrames together based on a shared column, similar to a JOIN in databases.
Apply	A Pandas method that lets you run a function on every row or column in a DataFrame.
Lambda	A short, one line function in Python written without giving it a name. Often used with apply.
NaN	Short for Not a Number. It represents a missing or empty value in a DataFrame.

Data Cleaning

Data Cleaning	The process of finding and fixing errors, missing values, duplicates, and inconsistencies in a dataset before analysis.
Missing Value	A blank or empty entry in a dataset where data was not recorded or was lost.
Null Value	Another term for a missing value. In Pandas, missing values appear as NaN.
Imputation	The process of filling in missing values with estimated ones, such as the mean or median of the column.

Duplicate	A row that appears more than once in a dataset with the same or very similar values.
Inconsistency	When the same information is recorded in different ways in a dataset. For example, writing a country as both UK and United Kingdom.
Data Type Error	When a column contains the wrong kind of data, such as a number stored as text, which stops calculations from working.
Standardisation	Making sure that all values in a column follow the same format, such as making all text lowercase or all dates the same shape.
Normalisation	Scaling numerical values to fit within a specific range, usually 0 to 1, so that different columns can be compared fairly.
Encoding	Converting categorical data into numbers so that it can be used in calculations or machine learning models.
Data Wrangling	The overall process of transforming and preparing raw data into a clean and usable format. Also called data munging.
Pipeline	A series of data cleaning or processing steps that run in a set order, one after the other.
Data Validation	Checking that the data in a dataset meets expected rules or formats before it is used for analysis.
Noise	Random errors or meaningless variation in data that can make analysis harder and results less accurate.

Data Visualization

Data Visualization	The use of charts, graphs, and maps to display data in a visual format that is easier to understand than raw numbers.
Bar Chart	A chart that uses rectangular bars to compare values across different categories.
Line Chart	A chart that uses a line to show how a value changes over time.
Scatter Plot	A chart that uses dots to show the relationship between two numerical variables.
Histogram	A chart that shows how often values fall within certain ranges. It is used to understand the distribution of a variable.
Box Plot	A chart that shows the spread and centre of a dataset using five summary values. It is useful for spotting outliers.
Heatmap	A grid where colours are used to represent the strength of values. Commonly used to show correlations between columns.
Pie Chart	A circular chart divided into slices to show how a whole is split into parts.
Axis	The horizontal or vertical line on a chart. The x axis goes left to right and the y axis goes up and down.
Legend	A key on a chart that explains what different colours or symbols represent.
Title	A label at the top of a chart that describes what it is showing.
Matplotlib	A Python library used to create a wide range of charts and graphs. It gives you detailed control over how your visualizations look.
Seaborn	A Python library built on top of Matplotlib that makes it easier to create attractive statistical charts with less code.
EDA	Short for Exploratory Data Analysis. The process of using charts and summary statistics to understand a dataset before modelling or reporting.

Trend	A general direction that data moves in over time, such as increasing sales or falling temperatures.
Annotation	Text or markers added to a chart to highlight specific data points or explain what something means.

Machine Learning Basics

Machine Learning	A branch of artificial intelligence where computers learn patterns from data without being explicitly programmed for every task.
Model	A mathematical representation built from data that can be used to make predictions or decisions.
Training Data	The portion of a dataset used to teach a machine learning model. The model learns patterns from this data.
Test Data	The portion of a dataset kept aside to evaluate how well a trained model performs on new, unseen data.
Supervised Learning	A type of machine learning where the model is trained on labelled data, meaning each record already has a known answer.
Unsupervised Learning	A type of machine learning where the model looks for patterns in data that has no labels or known answers.
Classification	A supervised learning task where the model predicts which category something belongs to. For example, spam or not spam.
Regression	A supervised learning task where the model predicts a numerical value. For example, predicting house prices.
Clustering	An unsupervised learning task where the model groups similar records together without being told what the groups should be.
Label	The answer or outcome column in a supervised learning dataset. It is what the model is trying to predict.
Prediction	The output of a machine learning model when it is given new data it has not seen before.
Accuracy	A measure of how many predictions a classification model got right out of all predictions made.
Overfitting	When a model learns the training data too well, including its noise, and performs poorly on new data.
Underfitting	When a model is too simple to capture the patterns in the data and performs poorly on both training and new data.
Feature Engineering	The process of selecting, transforming, or creating new columns in a dataset to improve model performance.
Algorithm	A set of rules or steps that a machine learning model follows to learn patterns from data.
Scikit Learn	A popular Python library for machine learning. It provides ready to use tools for classification, regression, clustering, and more.
Train Test Split	The process of dividing a dataset into two parts: one for training the model and one for testing how well it works.
Confusion Matrix	A table used to evaluate a classification model. It shows how many predictions were correct and where the model made mistakes.
Cross Validation	A technique where the data is split multiple times and the model is tested on different portions each time, to get a more reliable performance score.